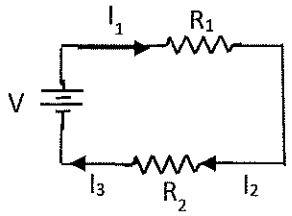
Worksheet # 4 Part A - Multiple Choice
Physics 1I

Name: Key Date: _____

Multiple Choice Questions 1 though 10

1) Two light bulbs are connected to a battery in series, which statement below is true about the current flowing through the circuit?



A) I_1 is larger than I_2, I_2 is larger than I_3
B) I_1 is smaller than I_2, I_2 is smaller than I_3
C) I_1 is larger than I_2 and $I_2 = I_3$
(D) $I_1 = I_2 = I_3$

2) Cathy has two resistors that she wants to connect in a circuit, so their equivalent resistance is the smallest possible, how does she connect them?

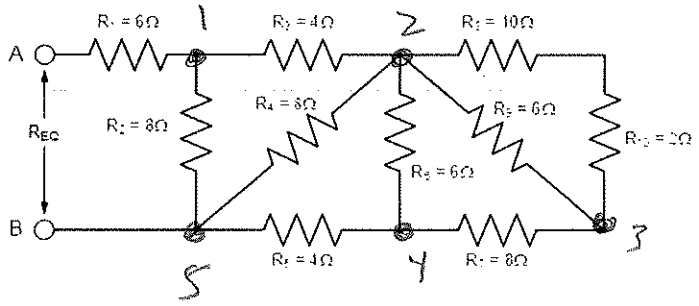
$R_{eq} = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}$ parallel

$R_p < R_s$

A) series.
(B) parallel.
C) either of these

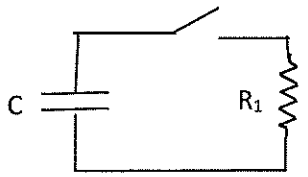
3) How many nodes are in the circuit below?

5



4) A fully charged capacitor at 100 V is wired in series with a resistor and switch. Determine the voltage across the capacitor as it discharges when the switch is closed for 2 seconds.

$\tau = RC = 10s$



C = 50 μC
R_1 = 200,000 Ω

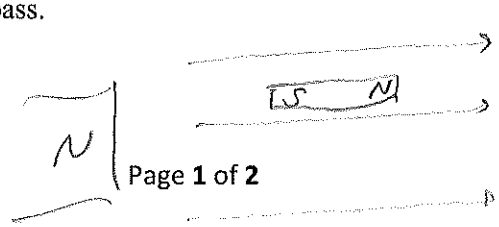
$V(t) = V_0 e^{-t/\tau}$

$V(t) = 100V(e^{-2s/10s})$

$V(t) = 82V$

5) Which magnetic pole of a compass needle points to a south magnetic pole of the Earth?

(A) north magnetic pole of the compass.
B) south magnetic pole of the compass.
C) both of these
D) none of these



$\vec{B}_{earth}$

S

6) A Sponge Bob magnet sticks to a refrigerator door because

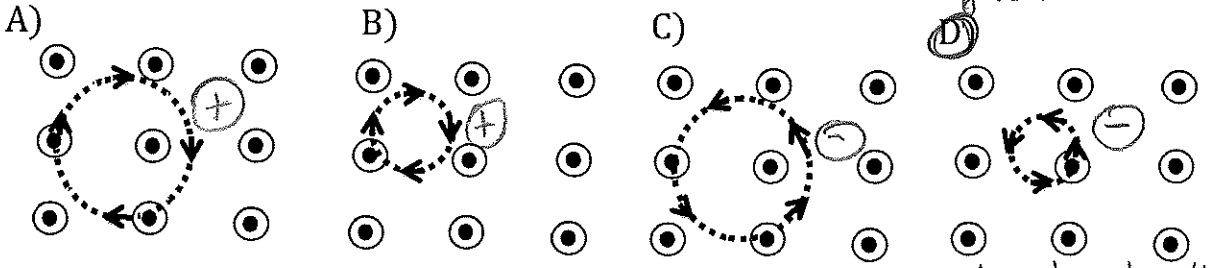
A) the magnet is electrically charged, and it induces a charge in the iron in the refrigerator.
B) the iron in the refrigerator is manufactured to be a north pole and the south pole of the magnet sticks to it.
C) The magnet induces the outer electrons of the ferromagnetic material in the refrigerator door to align their spin in the same direction creating the attraction.
D) The magnet induces a current in the iron that creates a magnetic field that attracts the magnet.
E) Sponge Bob sticks to anything.

7) Which field (electric or magnetic) can cause a moving charged object to change the magnitude of its velocity?

A) only an electric field
B) only a magnetic field
C) either an electric or magnetic field
D) none of these

8) Four objects with the same magnitude of net charge are shown below moving in a uniform magnetic field, which object has a net negative charge and has the smaller mass?

$r = \frac{mv}{|q|B}$

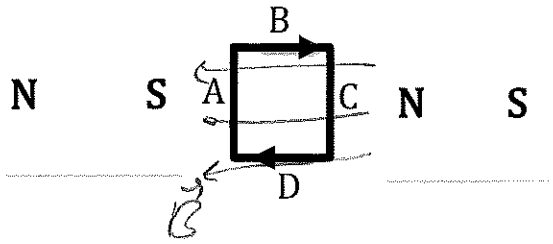


counterclockwise (−)
Smaller r → smaller m

9) A rectangular coil, with lengths labeled ABCD is placed between two magnetics as shown. If there is a current, *I* is flowing through this coil, what is the direction of the force acting on length C of this coil?

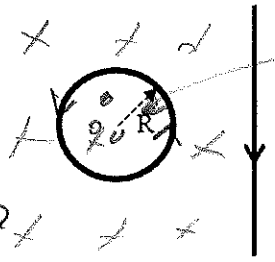
RHR #1

A) out of the page.
B) into the page.
C) to the right.
D) to the left.
E) there is no magnetic force on the wire.



10) A single current-carrying circular loop of radius *R* is placed next to a long, straight wire, as shown in the figure below. The current in the wire points down towards the bottom of the page and is of magnitude *I*. In which direction must the current flow in the loop to produce <u>zero magnetic</u> field at its center?

A) counterclockwise
B) clockwise
C) It could be either clockwise or counterclockwise.
D) The current in the loop must be zero.



$\vec{B}_{Loop}$ must be out of page
Therefore RHR #2
CCW Current

B_{wire} into page